APSIT SMART PARK

Problem Statement

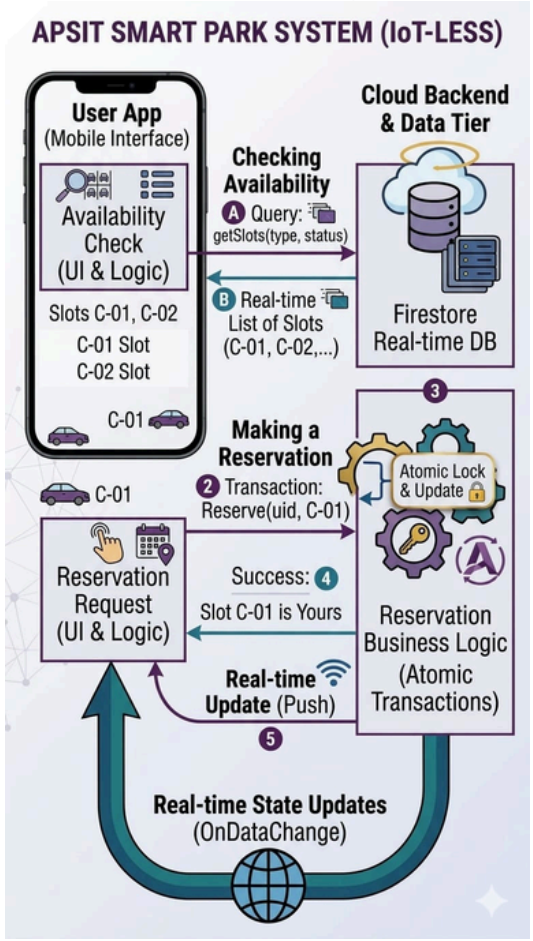
• Difficulty in finding available parking spaces for cars and bikes inside the APSIT campus, leading to time wastage and congestion.
• Unorganized parking and vehicles parked in wrong areas due to lack of proper monitoring and information about available slots.

Objectives

1.To optimize parking space utilization through a slot-based digital parking system.
2.To enhance parking discipline.
3.To implement user-linked slot allocation and restricted parking zones.
4.To integrate real-time monitoring and digital management for efficient campus parking operations.

Workflow

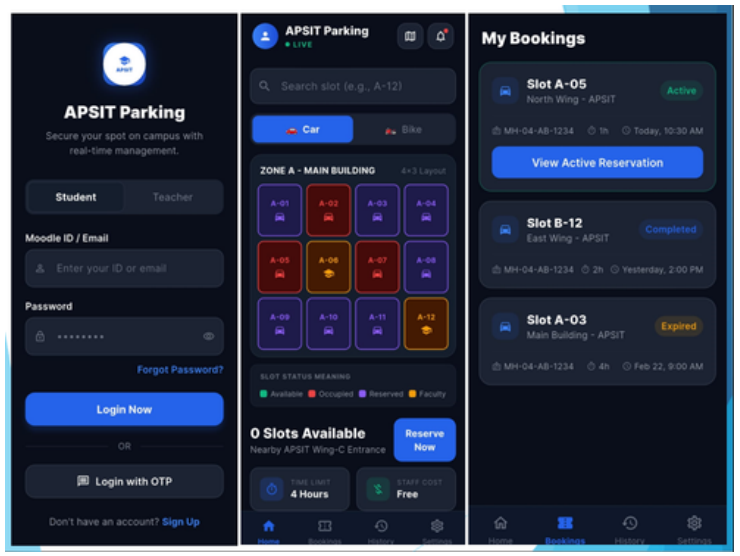
Block Diagram



Result

🚗 Parking System: Before vs After APSIT SMART PARK

Aspect	Before APSIT SMART PARK	After APSIT SMART PARK
Slot Availability	Manual checking	Real-time slot visibility
Time Wastage	High (searching for parking)	Low (pre-book slots)
Booking System	No proper system	Online slot booking
Traffic Inside Campus	More congestion	Smooth vehicle flow
Transparency	Unclear availability	Clear status (Available/Occupied)
Record Keeping	Manual / none	Digital history tracking
User Experience	Frustrating	Easy & convenient
Efficiency	Low	High
Security	Basic	Login-based controlled access



Key References

- Real-Time Slot Management: The primary reference for the project is the implementation of an interactive parking map that displays real-time availability for both car and bike slots within the APSIT campus.
- Secure User Accountability: A fundamental reference point is the use of the APSIT Moodle ID for secure authentication, ensuring that every reservation is linked to an authorized student or staff member.
- Automated Enforcement Logic: The system is designed to enhance campus discipline by utilizing color-based segregation for teacher and student zones and providing automated alerts for vehicles parked incorrectly.

Conclusion

APSIT SMART PARK is a software-based solution for managing campus parking efficiently. Built with Flutter and Firebase, it offers secure Moodle login, real-time slot tracking, and digital reservations. Features like color-coded zones and wrong-parking alerts improve space usage and discipline, creating a smart and organized parking system for APSIT.